including *NumPy* v1.19.5 for numerical computations, *stream-learn* v0.8.16 for handling data streams, *Sklearn* v1.0.2 for machine learning tasks [43], and the *Persim* v0.3.6 package for operations related to persistent homology [39, 44, 45]. The source code of our investigation and the datasets are available in the git repository[2].

4.4 Results

Figure 4 illustrates the latent space generated by the SOM projections of the data in case C. The curve (blue dots) in the figure represents the mean distance between the projected points and the mass centers of the clusters. It shows the evolution of the mean values for each of these distance matrices. In addition, Figure 4 shows the results of applying Pruned Exact Linear Time (PELT) algorithm [46] to detect changes in the sequence of these mean values. The PELT method is recognized for its computational efficiency, when compared with other change-point detection techniques [47]. This experiment was done offline with the purpose of visualizing the complexity of the problem. The aim was to show that detecting shifts in the generated data is not sufficient by merely applying a DR method and computing the mean distance to the mass centers. We consider Figure 4 as an illustration of the data stream characteristics for case study C. This figure also shows the detected change points using the PELT technique. The background colors represent the changes detected by PELT, and the vertical green lines indicate the injected shifts. The visualization using PELT serves as a representation of the problem's complexity. It was conducted independently of the other experiments, in which we compare the approaches using SOM, PCA, and Kernel-PCA. In other words, we are not comparing PELT to the other techniques; this experiment was conducted to evaluate whether the generated data presents difficulties for a well-known technique for change point detection. The other experiments simulate a continual learning environment, making them even more complex than when the problem is addressed offline. Figure 6 presents the results for the three DR techniques. These techniques were evaluated using case study C. The data stream was split into chunks of 250 samples. The vertical lines (represented by green dashed lines) indicate the injected shifts. The horizontal line represents a p-value of 0.05. According to the results, the linear projection is unable to accurately predict the shifts. This observation is consistent with other studies in the literature that discuss the limitations of linear projections in detecting distribution shifts [25, 35]. The performance of SOM appears to be slightly better to that of Kernel-PCA. For instance, refer to the p-values in the chunks between 30 and 35. Chunk size is a crucial parameter for experiments conducted in online settings. The influence of the chunk size is shown in Figures 7, 8, and 9, as well as in Table 1. Figure 7 shows the results of using linear projections before the analysis of the persistent entropy. There are two graphs: the top graph shows the results for chunks with 50 instances, and the bottom graph shows the results for chunks of 100 samples. This figure also highlights the limitations of linear projections for solving this specific problem, as the method provides few alarms and detects only a low number of drifts. Figure 8 presents the results of Kernel PCA for chunks with 50 and 100 samples. Similar graphics are depicted in Figure 9 where the results of SOM for chunks with 50 and 100 samples are presented. Table 1 summarizes the results of our experiments, including an evaluation of the impact of chunk size. The last two columns show the flags generated by the model using p-values with significance levels of 0.05 and 0.1. Additionally, we show the number of injected drifts in the experiment. Note that this number is an approximation due to the anomalies that can exist in the datasets (digits with a different number of holes than expected). According to the table, it seems that SOM may provide results using p-values with a 0.05 level of significance, while Kernel-PCA obtains better results when a p-value with a 0.1 level of significance is considered. As one might intuitively expect, smaller chunks decrease the quality of geometric pattern analysis. In contrast, larger chunks can encompass more than one shift. The optimal chunk size should be determined through experimentation, taking this trade-off into account.

5 Discussion

As far as we know, this is the first work that attempts to define drifts, including sudden changes in the topological features of the data. Our research hypothesis was to investigate whether the use of TDA may be helpful or not for

[2] https://github.com/sebabaster/Drift-persistence

monitoring topological changes and detecting "essential" differences between chunks of objects. In other words, if we have a data stream with different types of doughnuts and then suddenly the data contains coffee cups, a "traditional" drift detector would detect a drift (using the standard notion of drift [1]). However, if we consider purely topological information, both shapes (coffee cups and doughnuts) are considered equivalent. Therefore, it would not be appropriate to consider this type of changes as a drift; in the case that we are interested in detecting "essential" differences. Shapes with the same Euler characteristic cannot be considered as "essentially" different. The inclusion of TDA in the concept drift domain may be highly beneficial for avoiding false alarm when the data has only suffer simple transformations such as scaling, distortions and rotations.

To answer this research question, we implemented a set of experiments based in the MNIST dataset [41]. We defined three case studies based on changes in the Euler characteristic of the data. The digits were not individually visually inspected; instead, they were sampled using a uniform distribution. Consequently, there may be instances where an image was expected to have a specific number of holes, but the sample shows a different number. We illustrate these anomaly examples in Figure 5. The figure contains four graphics. The first image has at least two holes, whereas the expected number is one. The second image has one hole, but it represents either the number three or five, both of which typically have zero holes. The third image represents the number six, but it was not properly drawn, so the circle is not closed. Finally, we also show an image with pixels that are not connected, forming more than one connected component. Therefore, the analyzed scenarios contain noisy marks in the stream. For further analyses, it would be beneficial to generate new datasets with higher quality annotated data streams. In this study, we did not evaluate different strategies to mitigate the catastrophic forgetting problem. When a drift is detected, it is necessary to define an appropriate strategy to fine-tune and update the framework. This is especially important for SOM. We left this evaluation for future research. Another limitation of our experimental analysis is that the framework is composed of two modules: dimensionality reduction and persistent entropy. Further investigation is required to analyze the relevance of each of these modules and how they affect drift detection. Additionally, an ablation study considering the different steps in the pipeline would enhance the reliability of our experiments.

Last but not least, defining concept drift solely based on changes in topological features appears to present an incomplete understanding of the phenomenon. Analyzing objects only based on their topological aspects ignores the meaning of the objects. Human interpretation, including contextual insights and semantic understanding of the information, may also be relevant in some problems. For example, 6 and 9 are different digits according to semantic understanding, but a simple rotation can transform one digit into the other. Therefore, to define drift ignoring human interpretation, which is the case of purely consider topology also have limitations. Combining the three aspects–statistics and geometry (both considered in previous literature), and topology (the approach introduced here)–in the definition of drifts could be a promising direction. An additional interesting angle would be to also integrate the meaning of the information. Therefore, there are several avenues for future exploration, presenting many open questions that remain to be addressed.

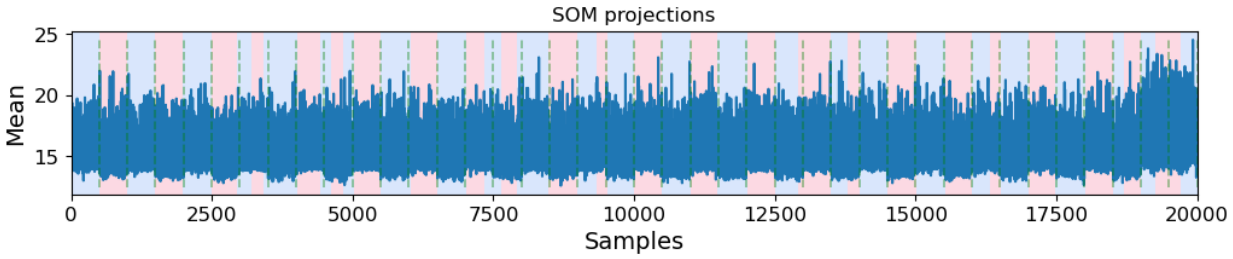

Figure 4: Example of the latent space. Off-line analysis of the latent space generated by the SOM projections, and applying change point detection over the distance matrix.

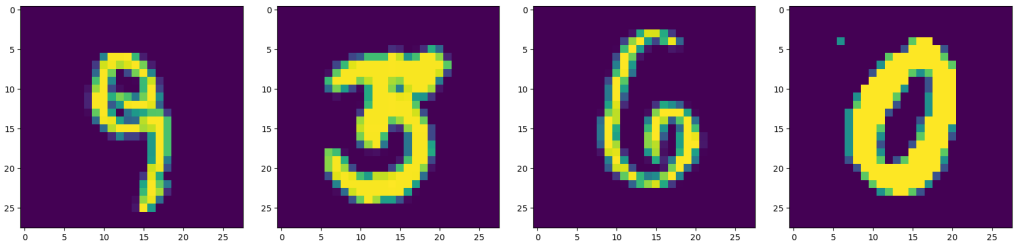

Figure 5: An example of digits with a structure different from the one assumed in the case studies. The first image has two holes instead of one. The second image, which can be a number 3 or 5, has a hole. The third image doesn't have any holes. The last image is not a connected component (it has an isolated pixel in the top-left).

6 Conclusions and future work

We introduced a novel approach to concept drift detection, leveraging algebraic topology and persistent entropy. We broaden the scope of concept drift, which is typically associated solely with statistical distribution changes in incoming data. In the explored approach, we also integrate the definition of drift to detect significant changes in topological features. The framework uses the SOM algorithm to transform the input space, reducing its dimensionality while preserving its topological characteristics. Then, the analysis of data drifts is conducted in the latent space. We explore the potential of persistent entropy to identify significant differences among data from consecutive chunks. We showed the performance of the method over three case studies (based on the MNIST dataset), and we compared the performance with PCA and Kernel-PCA. The proposed method does not make any assumption about the data distribution. Additionally, it can be applied to both supervised and unsupervised problems. We believe that this work is an initial step towards applying TDA in the area of concept drift. A potential direction for future research could involve evaluating the framework with different types of data streams, including dynamic graphs. Furthermore, it would be interesting to compare persistent entropy with other measures.

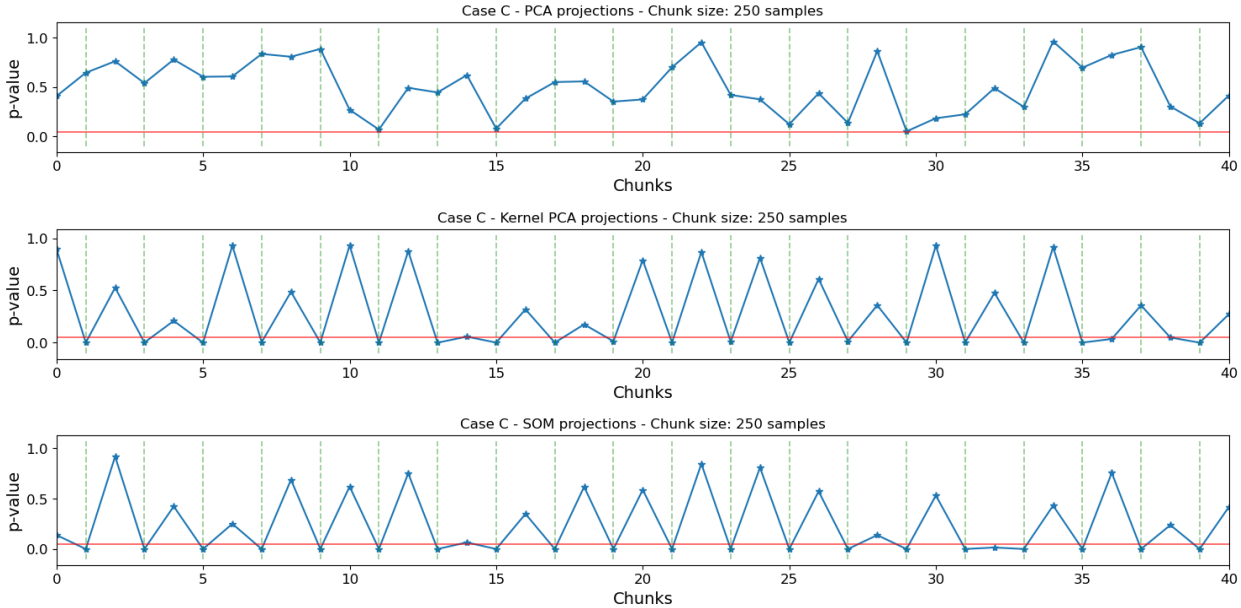

Figure 6: Comparison among the three projections. The comparison was made over the dataset C with chunk size of 250 samples.